

Notes: Boot, Milbourn & Schmeits (RFS, 2006)

Credit Ratings as Coordination Mechanisms

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1 Big picture

- **Question.** What economic role can credit ratings play if markets already process information and if rating announcements often appear to follow public information?
- **Core answer.** Credit ratings can act as **coordination mechanisms**. They help investors coordinate on the good equilibrium in settings where investor beliefs and firm risk choices are self-fulfilling.
- **Main friction.** Firms can choose either a viable low-risk project or a higher-risk project. Investors cannot observe the firm's project choice or recovery effort. Their beliefs affect debt pricing, and debt pricing affects the firm's project choice.
- **Multiple equilibria.** For medium-quality firms, if investors expect the viable project, they charge a low repayment and the firm chooses the viable project. If they expect the high-risk project, they charge a high repayment and the firm chooses the high-risk project.
- **Credit rating as focal point.** If enough institutional investors condition their investment decisions on the rating, then the rating can coordinate market beliefs and financing terms.
- **Credit watch as implicit contract.** The credit watch procedure is not just a warning label. It creates an implicit contract in which the firm can commit to recovery effort, and the rating agency monitors whether recovery effort is effective.
- **Price impact implication.** Downgrades after a credit watch can be strongly negative because they reveal failed recovery effort. Confirmations after credit watch can be positive because they indicate that recovery succeeded.
- **Role of institutional restrictions.** Rating-based investment mandates make the CRA's focal-point role credible: if enough investors follow the rating, other investors rationally follow too.
- **Limits.** If moral hazard is too severe or recovery effort is too ineffective, ratings cannot rescue the good equilibrium. In such cases, bank finance or private debt may dominate public debt.
- **Takeaway.** Ratings need not be valuable only because they reveal new hard information. They can be valuable because they coordinate beliefs, discipline firm behavior, and make dispersed public debt investors act more like a monitoring coalition.

2 Incremental contribution of the paper

- **Beyond the information-disclosure view of ratings.** Earlier debates often asked whether ratings reveal information not already known by the market. This paper adds a different channel: ratings can matter even when they do not reveal much new information, because they coordinate investor beliefs.
- **Ratings as equilibrium-selection devices.** The paper formalizes a setting in which firms' risk choices depend on investors' expectations. This creates multiple equilibria. The rating can select the good equilibrium by becoming the common reference point for investment decisions.
- **Credit watch as an incentive mechanism.** A central contribution is to model credit watch as an implicit contract between the CRA and the firm. The CRA can monitor recovery effort and condition the final rating on its success.
- **Institutional investors as the implementation device.** The paper links the theoretical mechanism to real institutional investment restrictions. Rating-sensitive investors make the rating consequential and create the incentive compatibility of the implicit contract.
- **New empirical predictions.** The model explains why watch-list announcements, confirmations, and downgrades should have different price effects depending on the probability that recovery effort succeeds and the firm's initial credit quality.

- **Public debt versus bank debt.** The paper qualifies the standard view that banks have unique monitoring advantages. CRAs can add a weaker but economically meaningful monitoring element to public debt markets.
- **Explains mixed empirical evidence.** If rating changes sometimes reflect pure information and sometimes reflect equilibrium coordination or recovery failure, average event-study evidence can look weak or mixed. The paper explains why conditioning on credit watch status matters.

3 Model primitives and measurement

3.1 Project opportunities and cash flows

Each firm needs one dollar of external financing. It can choose between:

- a viable low-risk project, denoted VP;
- a high-risk alternative, denoted HR.

The viable project pays X_{VP} with probability $\tilde{p}$ and zero otherwise. The high-risk project pays X_{HR} with probability q and zero otherwise, with $X_{HR} > X_{VP}$.

The paper assumes:

$$p_B X_{VP} > q X_{HR} > 1,$$

so the viable project is first-best efficient even after a bad credit shock, while both projects have positive net present value.

Interpretation: The high-risk project has a larger payoff conditional on success but a lower success probability. It is privately attractive when debt pricing makes downside risk cheap for equityholders.

3.2 Credit quality and signals

The firm's viable-project credit quality starts at p_0 . At date 1, a public signal can change it to:

$$\tilde{p} \in \{p_B, p_0, p_G\}, \quad p_B < p_0 < p_G.$$

After a bad signal, the firm can exert costly recovery effort. Recovery succeeds with probability β and restores the viable project's credit quality to p_0 ; otherwise credit quality remains at p_B .

The recovery-effort cost is $c > 0$, and the paper assumes:

$$p_0 X_{VP} - c > p_B X_{VP}.$$

Interpretation: Recovery effort is socially valuable when successful, but investors cannot observe the effort or its success unless the CRA monitors through credit watch.

3.3 Debt financing and repayment terms

All external financing is debt. If investors believe the firm will choose the viable project, competitive pricing gives:

$$F^{VP}(\tilde{p}) = \frac{1}{\tilde{p}}.$$

If investors believe the firm will choose the high-risk project, competitive pricing gives:

$$F^{HR} = \frac{1}{q}.$$

Because $\tilde{p} > q$, the high-risk repayment is larger: $F^{HR} > F^{VP}(\tilde{p})$.

Interpretation: Funding costs depend on investor beliefs. This is what makes beliefs self-fulfilling: high expected risk raises the coupon, and a high coupon can make the high-risk project privately optimal.

3.4 Institutional investors and rating-linked mandates

A proportion α of investors are institutional investors whose investment decisions are tied to the rating. If the CRA assigns a high rating consistent with the viable project, these investors can invest and conjecture that the firm chooses VP. If the rating is low, they abstain.

Interpretation: Rating-linked mandates are not just passive constraints. In this model they are the enforcement technology that gives the rating its focal-point power.

3.5 Credit ratings and credit watch

The CRA assigns a rating $\rho(\tilde{p})$ that is increasing in expected credit quality. After a bad signal, the CRA can initiate a credit watch procedure. With credit watch, the CRA monitors the firm's recovery effort and learns whether it succeeds.

Interpretation: Credit watch transforms the rating from a passive assessment into an implicit monitoring contract. The threat of a low rating after failed recovery makes recovery effort more attractive.

4 Model equations and theoretical results

4.1 The benchmark project-choice condition

If the firm faces a repayment F , its expected payoff from the viable project is:

$$\tilde{p}(X_{VP} - F).$$

Its expected payoff from the high-risk project is:

$$q(X_{HR} - F).$$

The firm chooses VP if:

$$\tilde{p}(X_{VP} - F) \geq q(X_{HR} - F).$$

Interpretation: This inequality shows how funding costs alter incentives. A larger repayment F increases the value of risk shifting because equityholders keep upside but default in failure states.

4.2 Theorem 1: three credit-quality regions

Theorem 1 divides firms into three regions:

1. **Low credit quality.** If $\tilde{p} < p$, the firm always chooses HR, regardless of investor beliefs.
2. **Medium credit quality.** If $p \leq \tilde{p} < \bar{p}$, the firm chooses whichever project investors expect. This is the multiple-equilibria region.
3. **High credit quality.** If $\tilde{p} \geq \bar{p}$, the firm always chooses VP, regardless of investor beliefs.

Interpretation: The interesting region is the middle one. In that region, a rating can matter because it can coordinate beliefs on the viable project and therefore change real behavior.

4.3 Theorem 2: a critical mass of investors can select the good equilibrium

Suppose a fraction α of investors believes the viable project will be chosen, while the remaining investors may expect HR. The total repayment is a weighted average:

$$F^\alpha = \alpha F^{\text{VP}} + (1 - \alpha) F^{\text{HR}}.$$

The firm compares:

$$q(X_{\text{HR}} - F^\alpha) \quad \text{and} \quad \tilde{p}(X_{\text{VP}} - F^\alpha).$$

Theorem 2 states that for every medium-quality firm there is a critical fraction $\alpha^*(\tilde{p})$ such that if $\alpha > \alpha^*$, the firm chooses VP and all investors rationally expect VP.

Interpretation: A sufficiently large group of investors playing the good equilibrium lowers funding costs enough to make the good equilibrium self-confirming.

4.4 Corollary to Theorem 2

The critical fraction $\alpha^*(\tilde{p})$ is decreasing in credit quality $\tilde{p}$.

Interpretation: Better medium-quality firms need a smaller rating-sensitive investor base to sustain the viable project. Lower-quality firms need a larger coordination force.

4.5 Lemma 1: failed recovery returns the firm to risk shifting

After a bad signal, if no recovery effort is exerted or if recovery fails, credit quality remains $p_B < p$. The firm then chooses HR. If recovery succeeds, credit quality is restored to p_0 , where $p \leq p_0 < \bar{p}$.

Interpretation: Recovery is crucial because without it the firm enters the low-quality region where investor coordination cannot rescue the viable project.

4.6 Theorem 3: no credit watch

Without credit watch, the CRA cannot observe whether recovery effort was undertaken or whether it succeeded. If recovery effort is sufficiently likely to succeed, the firm may undertake it and the CRA assigns an average rating based on:

$$p_n = \beta p_0 + (1 - \beta)q.$$

But the required institutional-investor fraction is higher than in the benchmark: $\alpha_n > \alpha^*$.

Interpretation: Without credit watch, the CRA is a passive coordinator. It can help only if recovery is sufficiently likely and if enough investors follow the rating. The rating cannot condition on the realized success of recovery.

4.7 Theorem 4: credit watch

With credit watch, the CRA observes recovery effort and its success. If recovery succeeds, the CRA assigns $\rho(p_0)$; if recovery fails or the firm does not exert effort, the CRA assigns the lowest rating $\rho(q)$.

The credit watch procedure increases the set of β values for which the firm undertakes recovery effort.

Interpretation: Credit watch has two effects. It improves coordination because successful recovery receives the precise high rating $\rho(p_0)$ rather than an average rating. It also improves incentives because failure is punished immediately by the lowest rating.

4.8 Theorem 5: good signals

If the firm receives a good signal and credit quality rises to p_G , the CRA assigns $\rho(p_G)$ and the firm chooses the viable project.

Interpretation: Good news pushes the firm out of the fragile multiple-equilibria region. The rating then confirms high credit quality rather than disciplining the firm.

4.9 Empirical predictions

The model generates five main predictions:

1. Bad public news lowers stock prices, but the negative effect is dampened when recovery is likely to succeed.
2. After credit watch is completed, confirmations are positive news and downgrades are negative news; the downgrade effect is related to the initial bad-news reaction.
3. Credit watch announcements can be positive or negative depending on market beliefs about recovery success.
4. Firms with lower initial credit quality experience faster deterioration after a bad shock.
5. The likelihood of being placed on credit watch is small for very low-quality firms, small for very high-quality firms, and large for intermediate-quality firms.

Interpretation: These predictions differ from a pure information-disclosure theory. They depend on the monitoring and coordination role of the credit watch process.

5 Identification and theoretical design

5.1 What the theory identifies

The paper identifies a mechanism rather than an empirical causal effect. The mechanism is:

rating-linked mandates $\Rightarrow$ belief coordination $\Rightarrow$ lower repayment $\Rightarrow$ viable project choice.

Credit watch adds:

CRA monitoring $\Rightarrow$ recovery incentives $\Rightarrow$ rating-dependent discipline.

5.2 Why ratings can matter without new information

If ratings merely repeat public information, they may still matter by shifting beliefs to a focal point. If enough investors condition on the rating, the rating changes the firm's financing cost and therefore its project choice.

5.3 Why institutional mandates are central

The model does not assume that every investor blindly follows the rating. It only requires a sufficiently large mass of rating-sensitive investors. Their behavior changes the equilibrium and gives other investors a reason to follow.

5.4 Why credit watch changes price effects

A downgrade after credit watch is not equivalent to a simple downgrade. It can reveal that recovery effort failed. A confirmation after credit watch can reveal that recovery succeeded. This explains why event studies should condition on credit watch status.

5.5 What the paper does not model

- It does not model CRA moral hazard or ratings shopping.
- It treats rating-linked investment restrictions as exogenous.
- It does not estimate the model empirically.
- It abstracts from strategic interaction among multiple CRAs except in the discussion section.
- It does not analyze welfare costs of excessive reliance on CRAs.

Interpretation: The paper is best read as a theory of why ratings can have a real role even if they are not pure information producers.

6 Tables, theorem blocks, and prediction blocks

The paper has no empirical figures. The main visual material consists of two timing tables plus theorem and prediction blocks. I place adjacent blocks together where possible to keep the notes compact.

6.1 Table 1: Sequence of events; Table 2: Sequence of events in the presence of a Credit Rating Agency (CRA)

Read:

- Table 1 presents the baseline model without an explicit CRA role. The firm raises financing, receives a credit-quality signal, may undertake recovery effort after a bad signal, and then chooses between VP and HR.
- The key baseline friction is that investors cannot observe the firm's project choice or recovery effort, so financing terms are based on beliefs.
- Table 2 adds the CRA. The CRA sets an initial rating, can initiate credit watch after a bad signal, monitors recovery effort, and assigns a rating used by institutional investors.
- The table clarifies why credit watch is an implicit contract: the firm is expected to take recovery action, and the CRA can condition the final rating on the outcome.

Economic message: The two tables show the incremental mechanism of the paper. Adding the CRA and credit watch turns a simple asset-substitution model into a coordination and monitoring model for public debt markets.

Table 1
Sequence of events

$t = 0$
• The firm announces that it needs to raise \$1 of external financing for investment • Nature determines the initial credit quality $\bar{p} = p_0$ of the viable project (VP)
$t = 1$
• Credit shock leads to $\bar{p} \in \{p_B, p_0, p_G\}$, with $p_B < p_0 < p_G$ (publicly observable) • Potential for costly "recovery effort" e of the firm. If successful, this can elevate p_B realization to p_0 • Actual financing and investment. Investors lend \$1 in exchange for a fixed repayment F', with $\tau \in \{VP(\bar{p}), HR\}$. The firm chooses between the viable project VP and the high-risk project HR
$t = 2$
• Project's payoff is realized. The firm makes debt repayment, if possible, and will default otherwise

(a) Table 1: baseline sequence of events

Table 2
Sequence of events in the presence of a Credit Rating Agency (CRA)

$t = 0$
• The firm announces that it needs to raise \$1 of debt financing to invest in a new project • Nature determines the initial credit quality $p_0 \in [p, \bar{p}]$ of the viable project VP • The CRA sets an initial rating ρ_0 based on the anticipated project choice, the viable project VP or the high-risk project HR
$t = 1$
• Both the firm and investors in the market may receive a publicly observable signal with respect to the credit quality of the viable project VP, leading to credit qualities $\bar{p} \in \{p_B, p_0, p_G\}$ • In case of a bad signal, the CRA could initiate a credit watch procedure, in which case it monitors the firm and specifies the (costly) recovery effort the firm needs to exert to maintain its initial rating ρ_0 (CRA—with credit watch) • The firm chooses whether or not to undertake recovery effort. The recovery effort, if undertaken, is successful with probability β. The CRA assigns a rating ρ, either not knowing whether recovery effort is undertaken (in absence of a credit watch) or after observing the recovery effort and its success or failure (with credit watch) • Institutional investors condition their investment decisions on the rating. If the CRA assigns a high rating reflecting project VP, then the proportion α of institutional investors is allowed to invest, and conjectures that the firm chooses the viable project VP. Otherwise, they abstain from investing. The remaining $(1 - \alpha)$ of investors form beliefs about the firm's investment behavior based on the size of α • Investors lend \$1 in exchange for a fixed repayment $F(\rho)$ based on their beliefs about the firm's investment behavior and the credit rating • The firm chooses between the viable project VP and the high-risk project HR
$t = 2$
• The project's payoff is realized. The firm will repay its debt obligation, if possible, and will default otherwise

(b) Table 2: sequence with CRA and credit watch

6.2 Theorem 1: Project choice by credit-quality region; Theorem 2: Critical investor threshold

Read:

- Theorem 1 is the foundation of the paper. It shows that the firm's project choice has three regions: always risky, belief-dependent, and always viable.
- Region 2 is where ratings matter. In this middle region, expectations determine debt terms, and debt terms determine project choice.

- Theorem 2 adds the coordination force: if a sufficiently large fraction of investors expects the viable project, the firm chooses the viable project and all investors rationally follow.
- The critical fraction α^* decreases with credit quality, so better firms need less coordination support.

Economic message: These results convert credit ratings from labels into equilibrium-selection devices. A rating matters because it can move the firm from a bad self-fulfilling equilibrium to a good self-fulfilling equilibrium.

the firm's equilibrium project choices are as follows. We focus on Bayesian Perfect Nash Equilibria.

Theorem 1. *The firm's project choice in equilibrium depends on its credit quality $\bar{p}$. In particular, the following regions can be distinguished:¹²*

1. *If $\bar{p} < \underline{p}$, then the firm always chooses project HR, regardless of investors' beliefs (anticipation) with respect to its project choice.*
2. *If $\underline{p} \leq \bar{p} < \bar{p}$, then the firm chooses whichever project investors anticipate will be taken. That is, if investors anticipate that the firm chooses the viable project VP, the firm optimally chooses VP. However, if*

(a) Theorem 1: project-choice regions, part 1

$$\Pr(VP \text{ succeeds}) \times (X_{VP} - F^\alpha) + \Pr(VP \text{ fails}) \times 0 = \bar{p}(X_{VP} - F^\alpha). \quad (5)$$

The critical proportion α^* equates (4) and (5). We can now proceed to our result that formalizes the impact of the proportion α of investors on the firm's equilibrium project choice.

Theorem 2. *For every $\underline{p} \leq \bar{p} < \bar{p}$, there exists a critical proportion of investors $\alpha^* = \alpha^*(\bar{p})$ that believe the viable project will be chosen, such that for $\alpha > \alpha^*$ the firm always chooses the viable project VP. In this case,*

(b) Theorem 2: critical mass of investors

6.3 Lemma 1: Failed recovery and Theorem 3: No credit watch

Read:

- Lemma 1 shows that if recovery fails after a bad signal, the firm is pushed into the low-quality region and chooses HR.
- Theorem 3 shows that without credit watch, the CRA cannot verify recovery effort. It must assign a rating based on expected recovery success rather than observed recovery.
- This passive CRA role requires a larger institutional-investor base to sustain the good equilibrium.

Economic message: Without credit watch, ratings can coordinate beliefs but cannot fully solve recovery-effort moral hazard. This is why the credit watch procedure is central rather than incidental.

or not to put in recovery effort, we first establish the following result.

Lemma 1. *If no recovery effort is put in, or the recovery effort fails (with a probability $1 - \beta$), the credit quality remains at p_B , with $p_B < \underline{p}$, and the firm will always deviate to the high-risk project HR. If recovery effort is undertaken and succeeds, then the credit quality of the firm is restored to p_0 .*

(a) Lemma 1: failed recovery

Theorem 3 (no credit watch). *In the case of a bad signal ($\bar{p} = p_B$), there exists a cutoff level $\bar{\beta} \in [0, 1]$ for the probability that recovery effort is successful, such that*

1. *If $\beta \geq \bar{\beta}$ (effective recovery effort), the firm undertakes recovery effort and the CRA assigns a rating $\rho(p_n)$, with $p_n \equiv \beta p_0 + (1 - \beta)q$, provided that the proportion of institutional investors that follow the rating exceeds α_n , with $\alpha_n > \alpha^*$, where α^* is defined in Theorem 2. The firm invests in the viable project if recovery effort is successful, otherwise it chooses project HR.*
2. *If $\beta < \bar{\beta}$ (ineffective recovery effort), the firm does not undertake recovery effort and the CRA assigns its lowest rating $\rho(q)$, reflecting*

(b) Theorem 3: no credit watch

6.4 Theorem 4: Credit watch and the corollary to Theorem 4

Read:

- Theorem 4 allows the CRA to monitor recovery effort and learn whether it succeeds.
- If recovery succeeds, the CRA assigns $\rho(p_0)$, which coordinates investors on the viable project.

- If recovery fails or the firm does not exert effort, the CRA assigns the lowest rating $\rho(q)$.
- The corollary says credit watch expands the range of recovery-success probabilities for which the firm undertakes recovery effort.

Economic message: Credit watch does not only convey information. It changes incentives. The firm exerts recovery effort because a failed or absent effort is punished by a low rating, and because successful effort receives a more precise rating.

anticipate the actual risk choice of the firm. The following theorem is the analog of Theorem 3 but now with the credit watch procedure.

Theorem 4 (credit watch). *In the case of a bad signal ($\bar{p} = p_B$), there exists a cutoff level $\underline{\beta} \in [0, 1]$ for the probability that recovery effort is successful, such that*

1. *If $\beta \geq \underline{\beta}$ (effective recovery effort), the firm undertakes recovery effort. The CRA assigns a rating $\rho(p_0)$ if the recovery effort is successful, provided that the proportion of institutional investors*

exceeds α^ , as defined in Theorem 2. The firm then chooses the viable project. If the recovery effort is not successful, the CRA assigns its lowest rating $\rho(q)$, and the firm chooses project HR.*

2. *If $\beta < \underline{\beta}$ (ineffective recovery effort), the firm does not undertake recovery effort, and the CRA assigns its lowest rating $\rho(q)$, reflecting the firm's choice for project HR.*

Corollary to Theorem 4. *A credit watch procedure increases the range of values of β for which the firm engages in recovery effort compared to the passive case with only holistic coordination by CRA: all firms with $\beta > \underline{\beta}$*

(a) Theorem 4: credit watch, part 1

(b) Theorem 4 and corollary, part 2

6.5 Predictions 1-2: Price effects of bad news and credit watch resolution

Read:

- Prediction 1 says bad public news lowers stock prices, but the fall is dampened when the probability of successful recovery is high enough.
- Prediction 2 says that completion of the credit watch procedure should create asymmetric effects: confirmations are positive; downgrades are negative.
- A downgrade after credit watch should be interpreted as failed recovery, not just as delayed publication of known bad information.

Economic message: These predictions explain why conventional event studies of rating changes can be hard to interpret. The same downgrade can mean different things depending on whether the firm had first been placed on credit watch.

Prediction 1. *Stock prices respond negatively to publicly-observed bad news, but the magnitude of this reaction is dampened the higher the likelihood of a successful recovery β , provided that this exceeds a critical minimum, that is, $\beta \geq \underline{\beta}$.*

The negative stock price reaction to bad news is muted in the presence of a credit watch because this procedure enables firms to attempt a restoration of their credit quality, and the market rationally anticipates this. This applies for firms with $\beta \geq \underline{\beta}$. For firms with $\beta < \underline{\beta}$, this can be observed even without a credit watch procedure.

Prediction 2. *Upon completion of the credit watch procedure, stock prices react positively to rating confirmations and negatively to rating downgrades. However, if the CRA downgrades the firm's debt after the conclusion of the credit watch, the negative stock market response that this*

Predictions 1-2: bad-news and credit-watch-resolution price effects

6.6 Prediction 3 and Predictions 4-5: Watch-list entry and deterioration dynamics

Read:

- Prediction 3 says a credit watch announcement can be either positive or negative, depending on how the market initially assesses the probability of recovery success.
- If the market thinks recovery is almost impossible, being put on watch is positive because it reveals that the CRA believes recovery is feasible.
- If the market thinks recovery is likely, being put on watch is negative because it reveals that monitoring is needed.
- Prediction 4 links lower initial credit quality to faster deterioration after a bad signal.
- Prediction 5 says watch-list placement is most likely for intermediate-quality firms: for very low-quality firms, watch is futile; for very high-quality firms, watch is unnecessary.

Economic message: The predictions make the model testable. The relevant empirical variables are not just rating changes, but the firm's initial quality, expected recovery probability, and whether the watch event is anticipated as good discipline or bad news.

participants only know the expected value of β . There we showed that only firms for which $\beta \in [\underline{\beta}, \beta_d]$ enter the credit watch procedure. However, market participants could have a high or low expectation of β , even when the true β lies in the range $[\underline{\beta}, \beta_d]$. Then our model extension suggests that the addition to the credit watch list could either dampen or exacerbate the negative announcement effect to the publicly observed deterioration of the firm's credit quality.

Prediction 3. *If the market assesses that the success of a firm's recovery*

(a) Prediction 3: watch-list entry

Prediction 4. *There will be an acceleration in the deterioration of credit quality for firms that face a drop in credit quality starting from a lower initial credit quality p_0 .*

This prediction would be consistent with the observed widening of credit spreads in uncertain times, as discussed in Section 2.5. An economic downturn could be interpreted as shifting downwards a firm's initial credit quality (p_0), thus limiting the effectiveness of the CRA and increasing the cost of debt financing. Finally, our model speaks to the likelihood of a firm entering the credit watch procedure in response to negative economic news, as opposed to being directly downgraded. This likelihood is nonmonotonic in the distribution of $\tilde{n}$.

(b) Predictions 4-5: deterioration and watch likelihood

7 Short summary

- The paper provides a theory of credit ratings as coordination devices.
- Firms face a moral hazard problem: after raising debt, they may choose a high-risk project rather than a viable low-risk project.
- Investor beliefs affect debt repayment terms, and repayment terms affect the firm's risk choice.
- For medium-quality firms, there are multiple equilibria: a good equilibrium with the viable project and a bad equilibrium with the high-risk project.
- If enough investors condition their investment decisions on the rating, the rating becomes a focal point.
- The CRA can coordinate investors on the good equilibrium, lowering funding costs and inducing the firm to choose the viable project.
- Credit watch is modeled as an implicit contract: the firm can exert recovery effort, and the CRA monitors and assigns a rating based on whether recovery succeeds.
- Credit watch expands the range of cases in which firms exert recovery effort.
- Downgrades after watch can be especially negative because they reveal failed recovery effort.
- Confirmations after watch can be positive because they reveal successful recovery and restored credit quality.

- The model explains why upgrades may have weaker average effects than downgrades.
- The model also explains why watch-list placement is most important for intermediate-quality firms.
- The paper adds a public-debt monitoring channel to the usual distinction between bank debt and public debt.

8 Conclusion

8.1 Core conclusion

Boot, Milbourn, and Schmeits argue that credit ratings can play a real economic role even when they are not purely superior information signals. Their key point is that ratings coordinate expectations. In a world with self-fulfilling risk choices, ratings can select the desirable equilibrium and reduce inefficient risk shifting.

8.2 Why this changes how we read rating announcements

The model implies that a rating change is not just an information event. It can also be the outcome of an implicit contract between the CRA and the firm. A downgrade following credit watch may reveal that recovery failed. A confirmation following credit watch may reveal that recovery succeeded. Therefore, the same formal rating change can have different economic meaning depending on its procedural history.

8.3 Importance of credit watch

The paper's most distinctive institutional insight is that credit watch is not a side feature of the rating process. Credit watch is the mechanism that gives the CRA a monitoring role. It lets the CRA observe recovery effort and lets investors condition on the CRA's final assessment. This makes the firm's recovery effort incentive compatible over a wider range of states.

8.4 Public debt versus private debt

The paper also refines the standard view that bank finance is uniquely suited to monitoring. Banks still have more direct control rights, but CRAs can partially reproduce monitoring in public debt markets through rating-sensitive investor mandates. Public debt can therefore contain an indirect monitoring mechanism when investors coordinate on ratings.

8.5 Policy and empirical implications

The theory has implications for rating regulation and empirical design. If ratings serve a coordination role, removing rating-based investment rules may reduce overreliance on CRAs but may also weaken a coordination mechanism that can discipline firms. Empirical work should separate: initial bad-news events, credit-watch announcements, credit-watch resolutions, rating confirmations, and downgrades after watch. Pooling these events can obscure the distinct mechanisms.

8.6 Limitations and open questions

The model treats rating-linked mandates as exogenous and abstracts from CRA moral hazard, ratings shopping, conflicts of interest, and the interaction among multiple rating agencies. These omissions are important, especially after later credit-market crises. However, the paper intentionally focuses on the prior conceptual question: why ratings can have value in the first place.

8.7 Takeaway

The paper's incremental lesson is that credit ratings can matter because they coordinate beliefs and enforce implicit recovery effort, not only because they reveal information. Ratings are therefore part of the governance structure of public debt markets. Their value comes from the interaction of investor mandates, firm incentives, and CRA monitoring.